

Replication package

Diversification, Growth, and Stability in Local Fishing Economies Kyumin Kim and Matthew N. Reimer, University of California, Davis

All results are built from four public CSV files:

```
./run_all.sh          # about 15 minutes  
RUN_BAYES=1 ./run_all.sh  # also runs the Bayesian check for Table S6 (about 20 minutes)  
STATA=/path/to/stata ./run_all.sh
```

The package ships with only code and raw data; `data/derived/`, `output/` and `logs/` are empty until you run it. The script runs the Stata stage, then the R stage. Figures are written to `output/figures/`, LaTeX table bodies to `output/tables/`, and saved estimates to `output/estimates/`. To run Stata alone, change to the package root and run `do code/stata/00_master.do`.

1. Requirements

Software	Notes
Stata 17 or later	<code>sfcross</code> , <code>parmesh</code> , <code>mat2txt</code> and <code>estout</code> are bundled in <code>code/ado/</code>
R 4.4 or later	<code>here</code> , <code>tidyverse</code> , <code>haven</code> , <code>sf</code> , <code>patchwork</code> , <code>viridis</code> , <code>rnaturalearthdata</code> , <code>isoband</code> , <code>plot3D</code> ; <code>rjags</code> and <code>coda</code> for Table S6
JAGS 4.3	Table S6 only. <code>rjags</code> 4-17 does not build against JAGS 5, so install 4.3.x (Homebrew currently ships 5)

On Apple Silicon, the CRAN binary of `rjags` may fail to find `libjags`. Build it from source:

```
install.packages("rjags", type = "source",  
  configure.args = "--with-jags-include=/opt/homebrew/include/JAGS --with-jags-lib=/opt/homebrew/lib/JAGS")
```

2. Data

All inputs are in `data/raw/` and are never modified.

File	Source
akfish-data-CFECpermits.csv	Alaska Commercial Fisheries Entry Commission permit records
employment-by-industry-community-and-year.csv	Alaska Local and Regional Information (ALARI)
employment-summary-data-by-community-and-year.csv	ALARI
Total_pop.csv	Alaska Department of Labor population estimates (population sheet extracted from the source workbook)

Study period 2000–2016; estimation sample 177 communities.

3. Exhibits

Exhibit	Produced by
Tables S1–S5	code/stata/22_tables.do, from the estimates saved by
Table S6	20_estimate_frontiers.do and 21_mrt.do
In-text MRT estimates	code/stata/39_panel_export_for_bayes.do, then code/R/14_tableS6_bayesian.R
Fig. 1	code/stata/21_mrt.do
Fig. 2	code/R/15_fig1_ak_overview.R
Fig. 3	code/R/13_fig2_sector_composition.R
Fig. 4, Fig. S2	code/R/11_fig3_frontier_concept.R (illustrative; not estimated)
Fig. 5	code/R/12_fig4_figS2_frontier_quantiles.R
Fig. 6	code/R/16_fig5_mrt_marginal_panels.R
Fig. S1	code/R/17_fig6_mrt_substitution.R
	code/R/10_figS1_diversification_trend.R

The manuscript in `paper/` reads its figures from `output/figures/` and its tables from `output/tables/`, so run the package before compiling it.

4. Structure

Script	Stage
<code>code/stata/01–12</code>	Build <code>master_local_fish_max.dta</code> from the raw CSVs (order matters)
<code>code/stata/20</code>	Estimates every frontier specification once and saves the estimation sample
<code>code/stata/21</code>	MRT, its elasticities, and percentile grids, from the saved spec 3 estimates
<code>code/stata/22</code>	Table bodies
<code>code/stata/30–39</code>	Two-period panel, used only for Table S6
<code>code/R/10–17</code>	Figures and the Bayesian check
<code>code/R/setup.R, config/</code>	Paths and shared packages

`code/ado/ehdf_prep.ado` builds the normalized variables and translog terms shared by the estimation scripts.

5. Notes

- The `keep if` in `20_estimate_frontiers.do` leaves 179 rows; `sfcross` drops those with a missing regressor, leaving 177 rows. Variables are normalized on the 179 rows, the frontiers are estimated once, and `estimation_sample.dta` (the 177 communities, with the normalized variables) is what every later script and all figures use.
- All MRT quantities (Tables S2 and S5, the elasticities, the cross-partial, and Figs. 5 and 6) are computed by the same analytic formula from the specification 3 estimates, with all other log regressors at zero (i.e., their mean).
- Table S6 uses MCMC. `set.seed(123)` is set, but JAGS is not bit-identical across platforms; posterior means and SDs should agree to the second or third decimal.
- `logs/` holds `00_master.log` and one log per R script.

6. Licence

Code is MIT, outputs and the manuscript are CC BY 4.0, the raw data follow the issuing agencies' terms, and the bundled Stata packages in `code/ado/` keep their authors' terms. See `LICENSE`.